

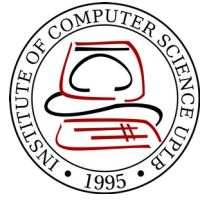
CMSC 137

Data Communications and Networking

Lecture Module
01 - Overview

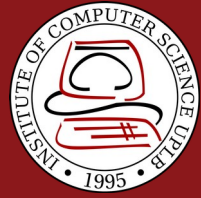


University of the Philippines
LOS BAÑOS



Module 1: Overview

Rozano S. Maniaol
Lecturer

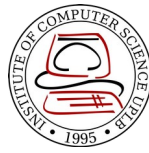


“What hath God wrought?”

Samuel F. B. Morse

Transmission of the first telegraph message on 24 May 1844 from Supreme Court chamber in Washington, D.C. to the B&O's Mount Claire Station in Baltimore, Maryland

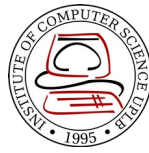
Introduction



Data communications and networking have changed the way we do business and the way we live

- Decision makers who require immediate access to accurate information can get it instantaneously
- Communities in the world are not isolated anymore

Terms



Communication – information sharing, local or remote

Telecommunication – communication at a distance

Data communication – exchange of data between devices via some form of transmission medium such as wire cable

Data – information presented in whatever form agreed upon by parties creating and using the data

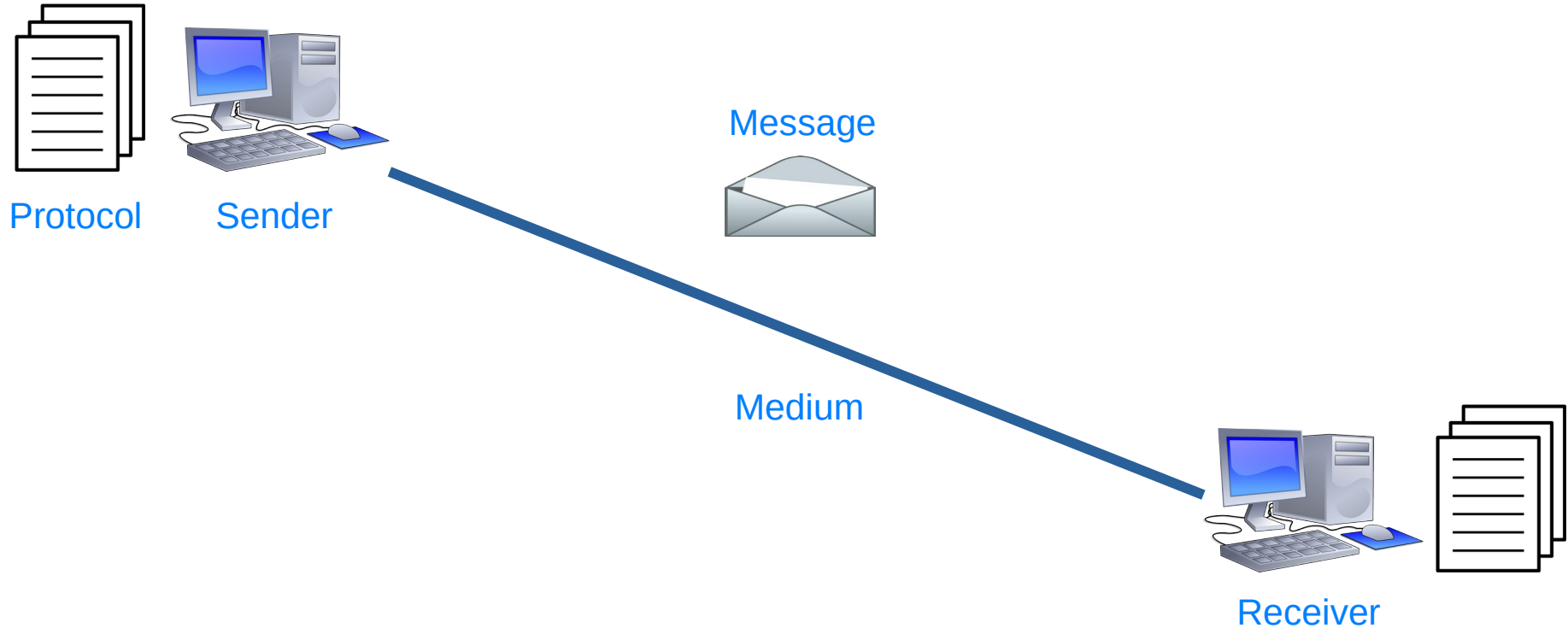
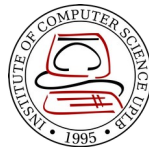
Communicating devices must be part of a data communication system (DCS), a combination of hardware and software

Effectiveness of a DCS

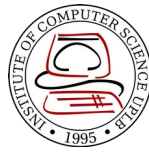


- **Delivery**
 - must deliver data to the correct destination
- **Accuracy**
 - must deliver data accurately
- **Timeliness**
 - must deliver data in a timely manner
- **Jitter**
 - variations in the packet arrival time

Components of a Data Communication System



Data Representation



Text

- bit pattern, sequence of 0's and 1's
- different set of patterns (code) is used to represent text symbols

Unicode – uses 32 bits to represent a symbol or character used in any language in the world

American Standard Code for Information Interchange (ASCII) – uses 8 bits and constitute the first 127 characters in Unicode

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BEL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	;	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16	10	10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[ENG OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	~
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	DEL
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	
32	20	100000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

Data Representation

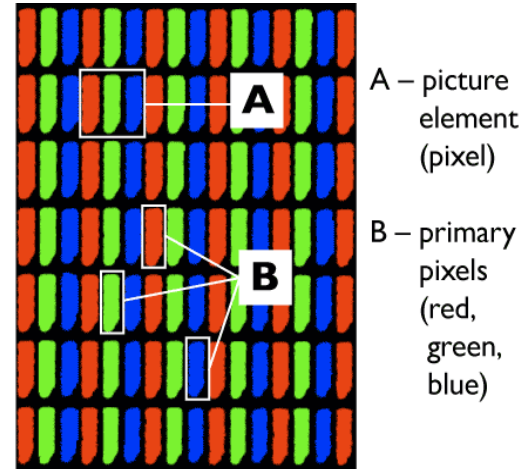


Numbers

- represented as bit patterns but is directly converted to a binary number
ex. Number 5 is represented as 00000101

Images

- bit patterns represented as a matrix of picture elements (pixels)



<http://courseweb.stthomas.edu/>

Data Representation



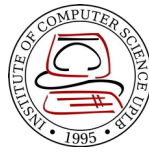
Audio

- recording or broadcasting of sound or music
- represented by a continuous signal

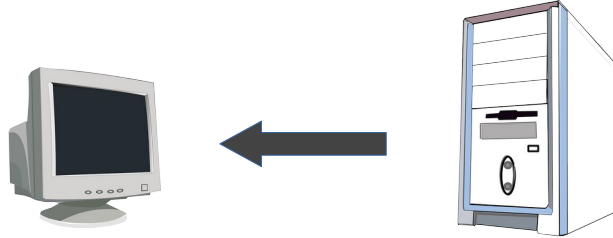
Video

- recording or broadcasting of a picture or movie
- can be
 - Continuous: via TV Camera
 - Discrete: via combination of images (frames)

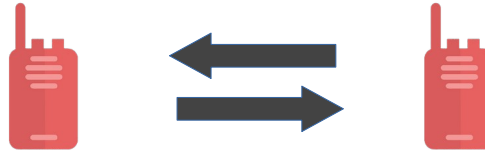
Data Flow



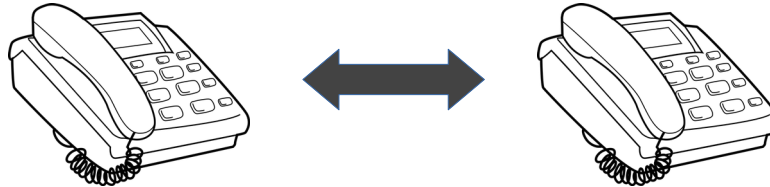
Simplex

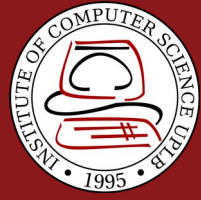


Half-duplex



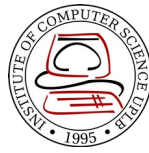
Full duplex / duplex



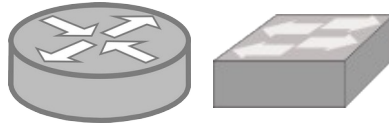


Network

Network



- Interconnection of a set of devices capable of communication
- Devices can be:
 - **Host** (end-system) like large computer, desktop, laptop, mobile phone, or security system
 - **Connecting device** like router, switch or modem



- Uses distributed processing
 - A task is divided among multiple computers

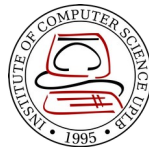
Network Criteria



Performance

- Can be measured in many ways, including:
 - **Transmit time** – amount of time required for a message to travel from one device to another
 - **Response time** – elapsed time between an inquiry and a response
- Depends on the *number of users, type of transmission medium, capabilities of connected hardware, efficiency of software*
- Ideal metrics: more **throughput**, less **delay**

Network Criteria

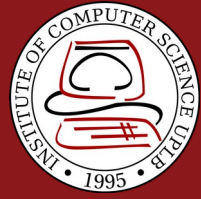


Reliability

- Measured by the *frequency of failure*, the *time it takes a link to recover from a failure*, and the *network's robustness in a catastrophe*

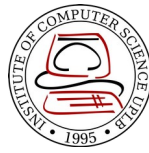
Security

- Protecting data from *unauthorized access*, protecting data from *damage*, implementing *policies and procedures for recovery from breaches and data losses*

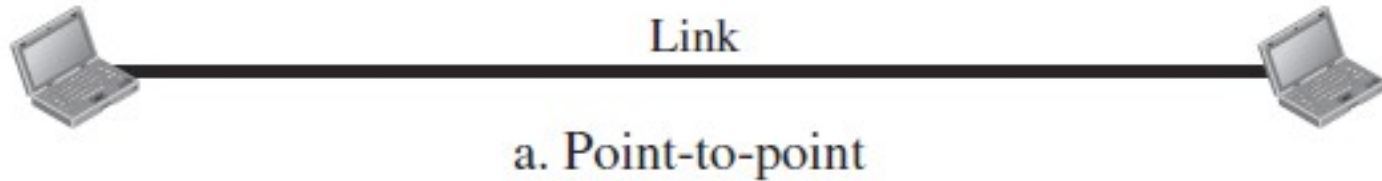


Network Attributes

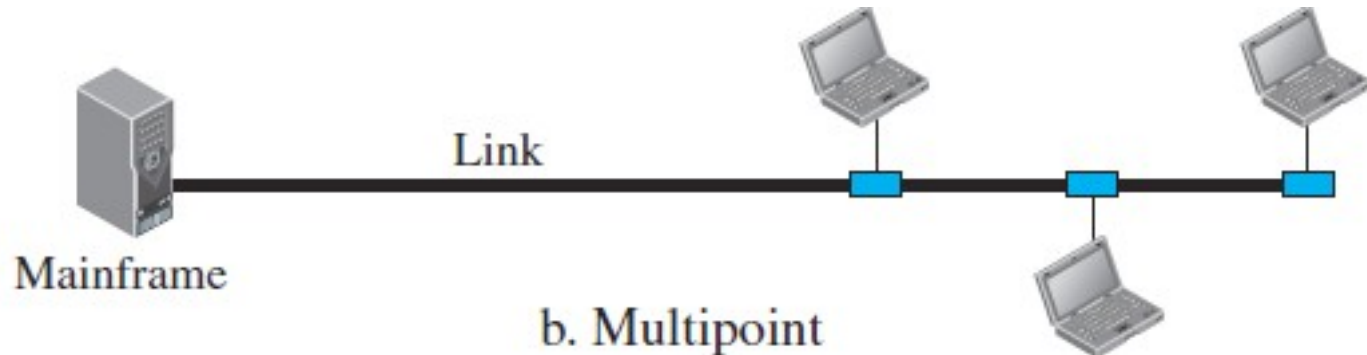
Physical Structure: Type of Connection



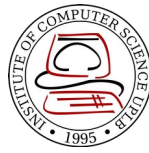
Point to Point



Multipoint / Multidrop



Physical Structure: Topology



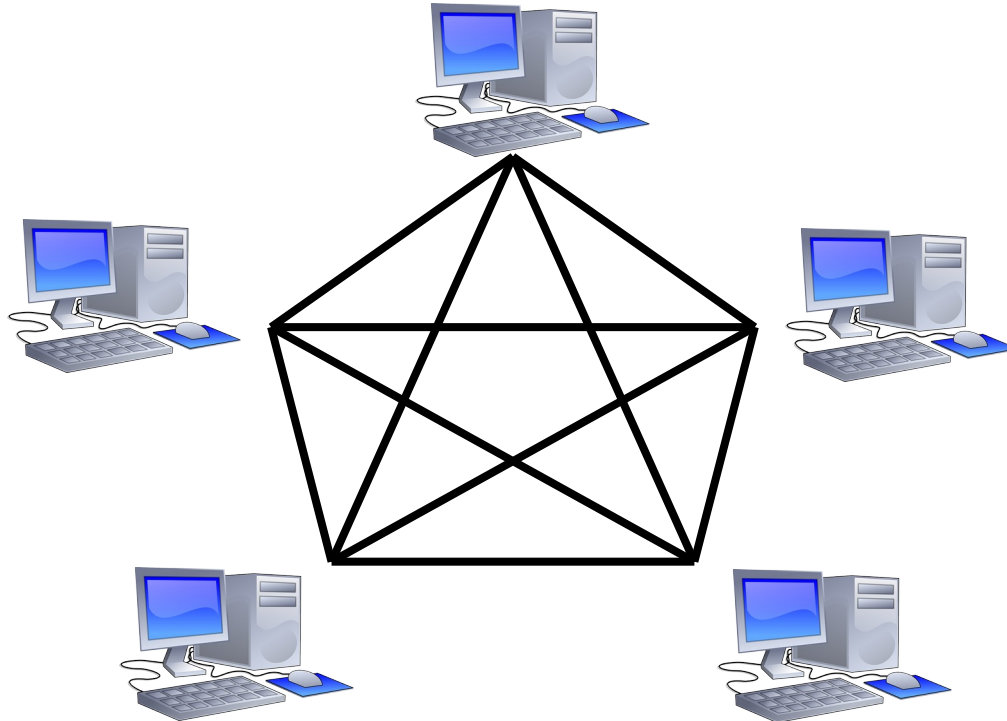
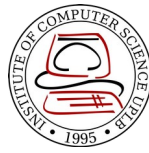
- Refers to the way in which a network is laid out physically
- Geometric representation of the relationship of all the links and nodes to one another
 - Mesh
 - Star
 - Bus
 - Ring

Mesh



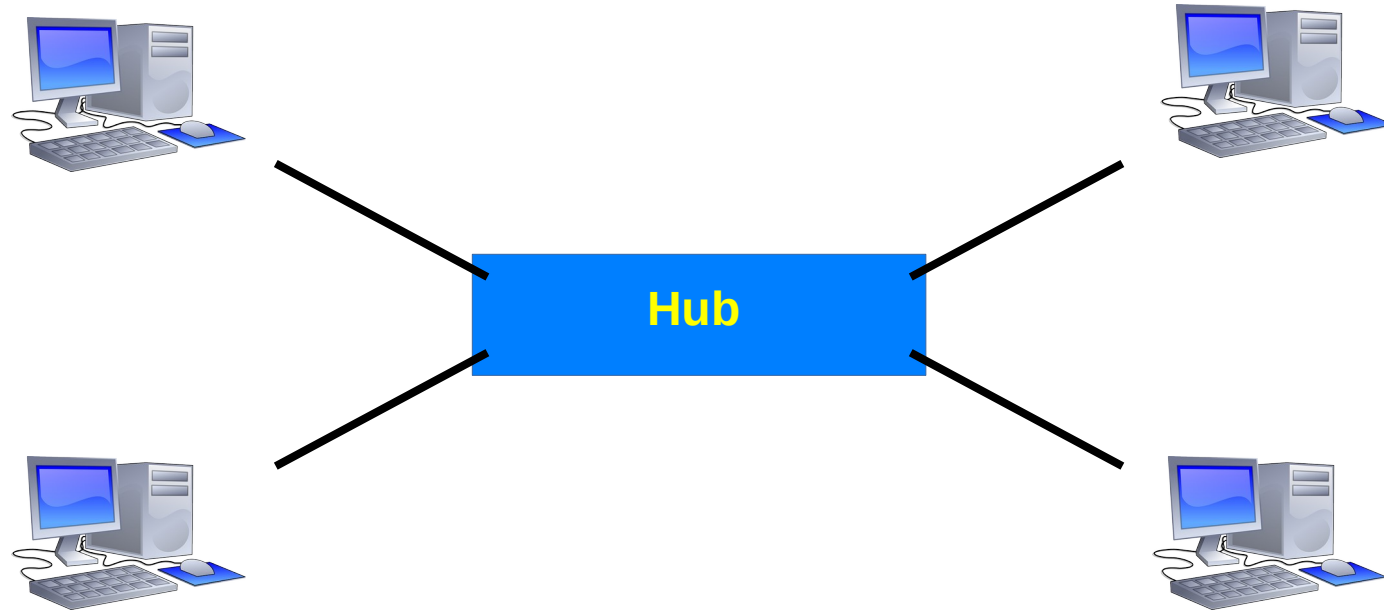
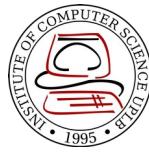
- Every device has a **dedicated point-to-point link** to every other device
- Number of duplex-mode links: $n(n-1)/2$
 - How many links for 5 nodes? $5(5 - 1) / 2 = \mathbf{10}$
- Advantages
 - No link sharing, robust, privacy or security, makes fault isolation easy
- Disadvantage
 - High cost
- ex. Regional telephone connections

Mesh

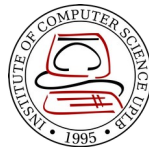


- Has a dedicated point-to-point link only to a central controller called a **hub**
- No direct traffic between devices, controller relays traffic
- Advantages
 - Cheaper than mesh, easy to install and reconfigure, robustness
- Disadvantage
 - Hub is the single point of failure, sometimes more cabling is required

Star

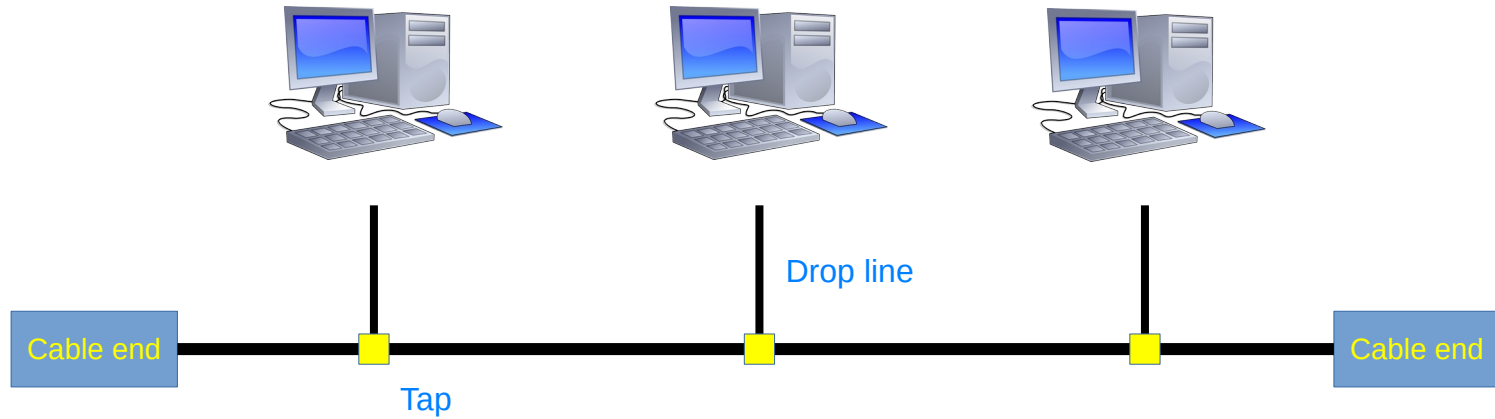
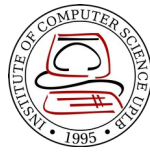


Bus

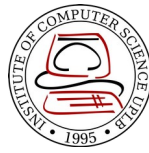


- Uses multipoint connection
- **Backbone cable** links all devices in the network
- Connection is via drop lines and taps
- Advantages
 - Ease of installation, cost less
- Disadvantages
 - Difficult reconnection and fault isolation, reduced quality because of signal reflection in taps, single break in backbone stops all transmission

Bus

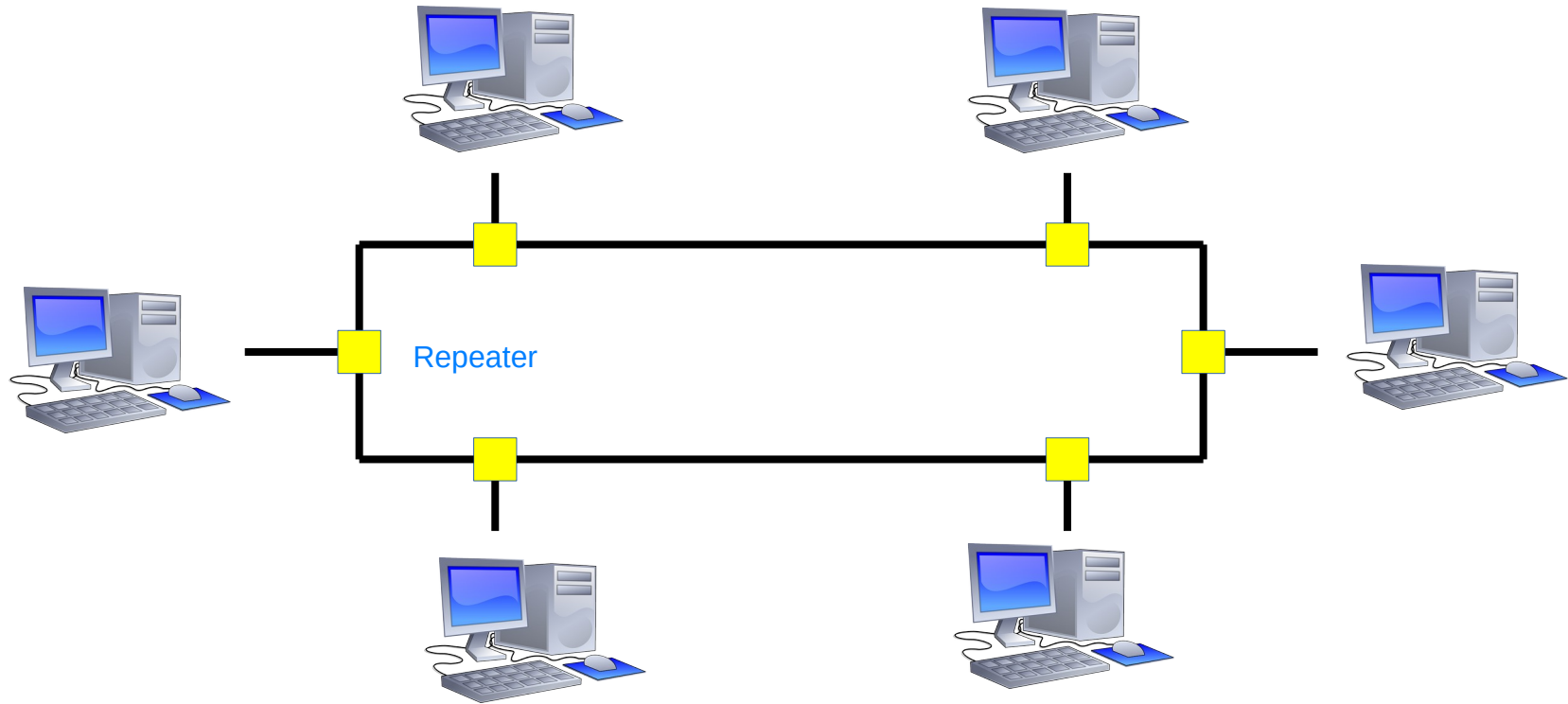


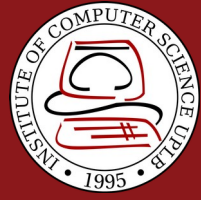
Ring



- Each device has a dedicated point-to-point link with **only two devices** on either side of it (physically or logically)
- Signal passed along the ring in one direction, from device to device, until it reaches its destination
- **Repeaters** regenerate the signal intended for another device
- Advantages
 - easy to install and configure, simplified fault isolation
- Disadvantage:
 - a break in the ring can disable the network (solved by using a dual ring)

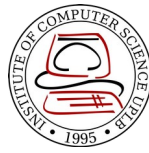
Ring



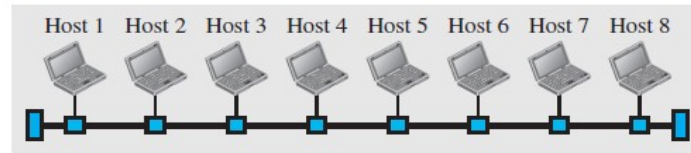


Network Types

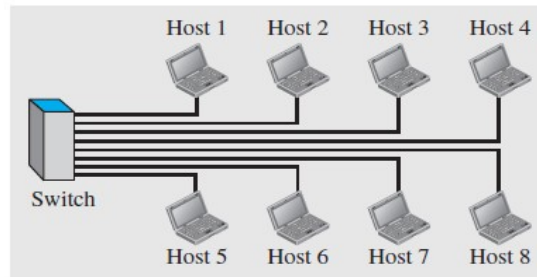
Local Area Network (LAN)



- Interconnects hosts
- Usually privately owned; can be as simple as two PCs and a printer
- Limited to a few kilometers, spanning an office, a building or campus

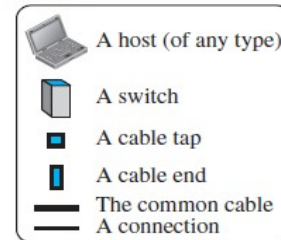


a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend

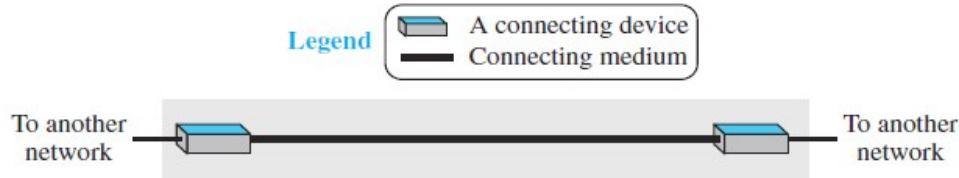


Wide Area Network (WAN)

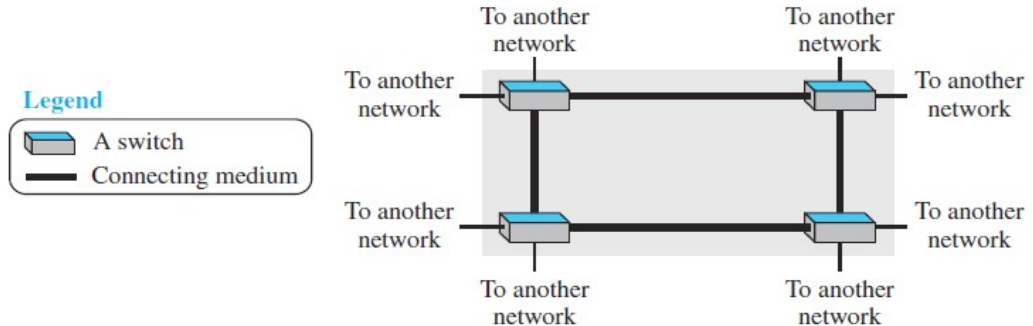


- Interconnects connecting devices
- Normally created and run by communication companies
- Long distance transmission of data

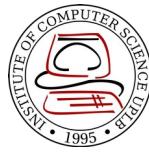
Point to Point WAN



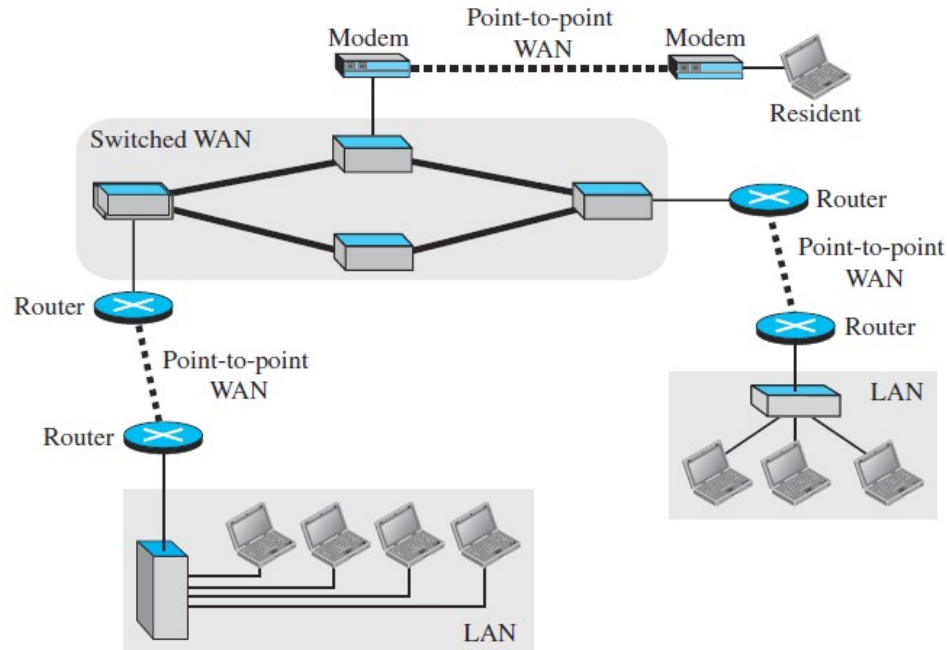
Switched WAN



internetwork



When 2 or more networks are connected

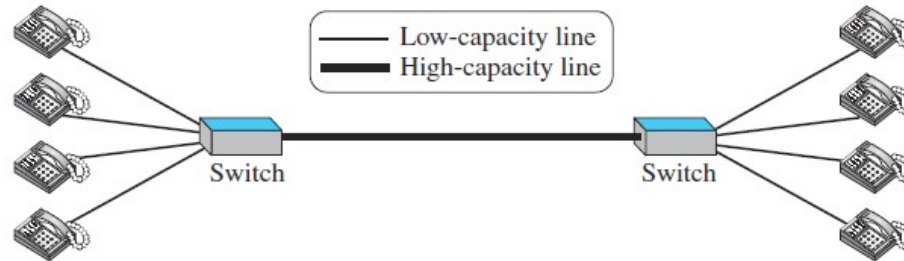


Switching

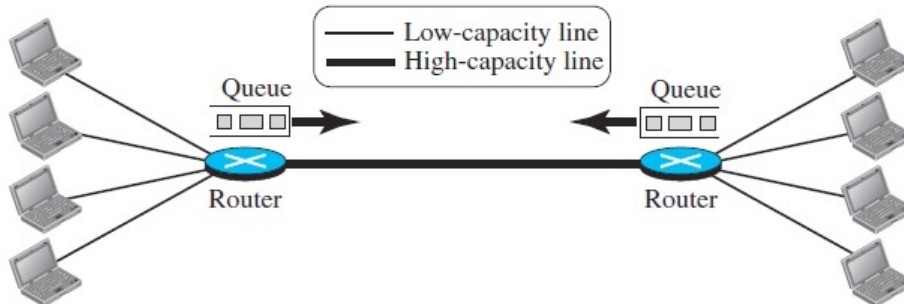


An internet is a switched network in which a switch connects at least two links together

Circuit-switched Network



Packet-switched Network



Reference



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.